

Cyclistic Bike-Share Case Study

Analyzing differences in how members and casual riders use Cyclistic bikes

Author: Sergey Kasatov

Google Data Analytics Professional Certificate Capstone

Date: July 2026

Slide 2: The Business Task & Objectives (Ask)

- Analyzing historical trip data to understand how annual members and casual riders use Cyclistic bikes differently.
- Using these key insights to design a targeted marketing strategy aimed at converting casual riders into annual members.
- Key Stakeholder Moreno (Director of Marketing) wants to design digital campaigns to maximize high-margin annual memberships.
- Final success requires backing up recommendations with professional data visualizations and clear statistical evidence.

KEY OBJECTIVES

Identify Core Behaviors

Analyze trip volumes, durations, and weekly patterns to draw distinct profiles of members and casuals.

Formulate Strategy

Convert casual riders into annual members by appealing to their specific usage patterns.

Data-Driven Recommendations

Backup findings with clear visualizations and statistical metrics for executives.

Slide 3: Data Prep & Processing (Prepare & Process)

- Data Source: Divvy public bike-share historical trip data from July 2025 to June 2026 (12 months).
- Initial Dataset: ~5.93 million rows loaded and merged using Python and pandas.
- Calculations: Computed ride length (ended_at - started_at) in minutes and extracted day of the week.
- Data Cleaning: Excluded trips shorter than 1 minute or with negative durations (docking checks and errors).
- Final Cleaned Dataset: 5,770,103 rows utilized for analysis (2.73% of erroneous rows removed).

DATA PIPELINE SUMMARY

Raw Trips Collected

5,932,349 rows merged from 12 CSV monthly files.

Standardized Datetimes

Parsed started_at and ended_at string columns to datetime objects.

Calculated Trip Duration

Derived ride_length in minutes to check trip lengths.

Filtered Dataset Size

5,770,103 clean rows kept for analysis (162,246 rows removed).

Slide 4: Key Insights: Volume & Duration (Analyze)

- Annual Members account for 64.7% (3.73M rides) of the total, while Casual Riders represent 35.3% (2.03M rides).
- Casual riders travel for an average of 22.57 minutes per trip, whereas Annual Members average 12.73 minutes.
- Casual riders ride almost twice as long per trip compared to members, strongly suggesting recreational and leisure usage.
- Members' trips are shorter and more uniform, which indicates routine utility-driven trips (commuting to work/school).

ANNUAL MEMBERS

64.7%

3.73 Million Trips

Represents the core user base. They ride mostly during weekdays for routine commutes.

CASUAL RIDERS

35.3%

2.03 Million Trips

Key target for conversion. They use bikes heavily for weekend recreation.

CASUAL TRIP DURATION

22.57 min

Average Trip Length

Casual trips are almost 2x longer than members, indicating leisure usage patterns.

MEMBER TRIP DURATION

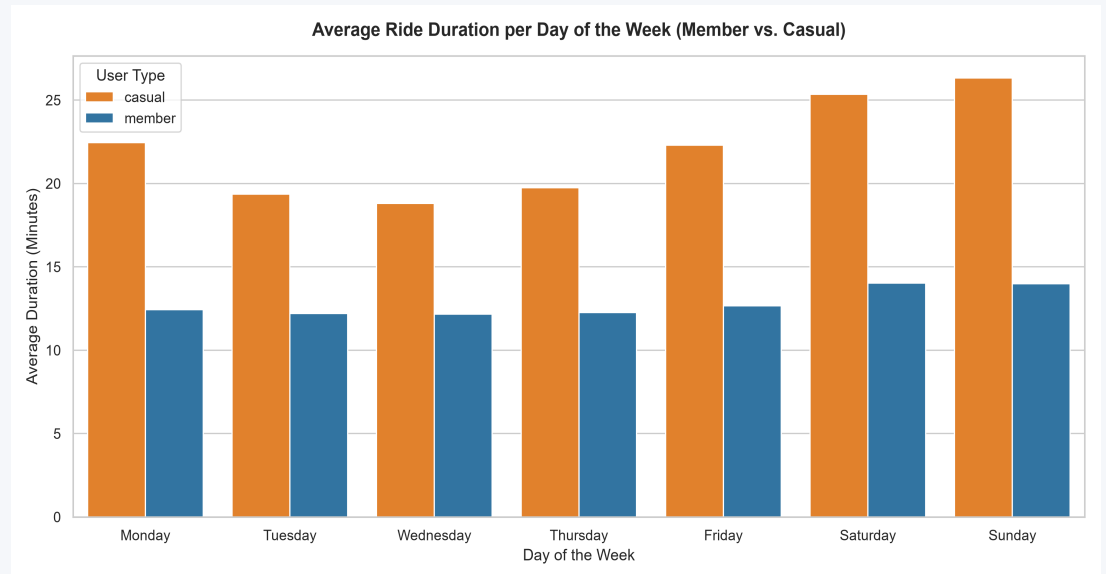
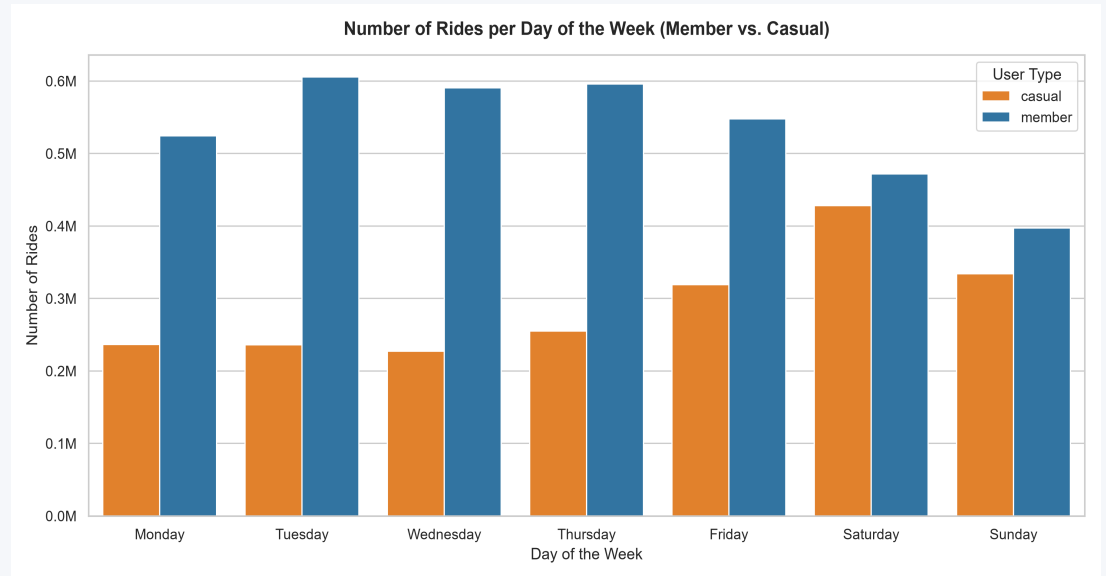
12.73 min

Average Trip Length

Shorter and more focused trip durations reflect quick utility-driven commutes.

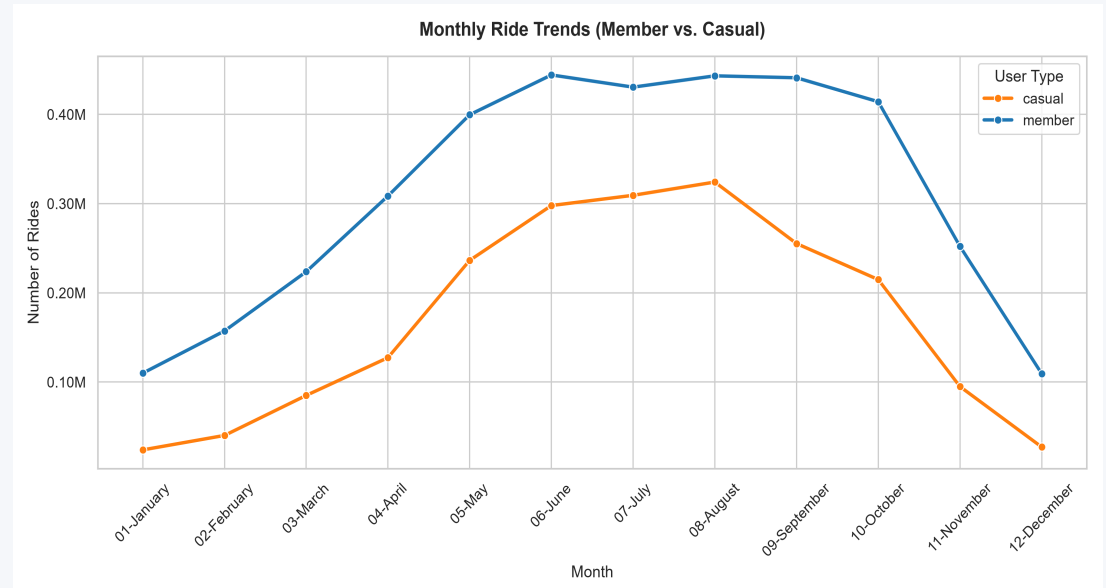
Slide 5: Weekly Patterns: Rides Count & Trip Duration

- Members show high activity on weekdays (Monday to Friday), reflecting commuting behavior.
- Casual riders peak heavily on weekends (Saturday and Sunday), reflecting leisure activity.
- Casual riders ride significantly longer than members every single day of the week, with durations peaking on weekends.



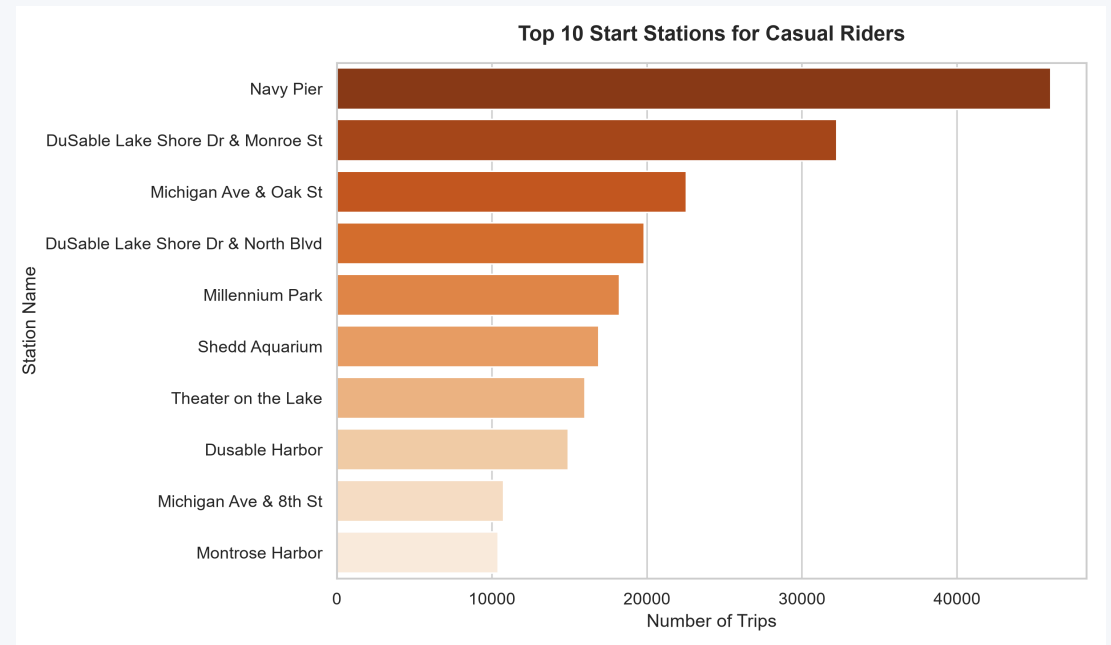
Slide 6: Monthly Seasonality Analysis

- Both members and casual riders exhibit strong seasonal patterns, with rides peaking in summer (June to August) and dropping in winter.
- Casual riders are extremely sensitive to seasons: summer rides are over 10x higher than winter rides.
- Members maintain a highly stable base of rides throughout the year (spring and autumn remain solid for work commutes).



Slide 7: Top Start Stations for Casual Riders

- Casual riders are highly concentrated around waterfront locations, parks, and major tourist hubs.
- Navy Pier is the absolute #1 starting location for casual riders with over 45,000 trips.
- Other popular stations include DuSable Lake Shore Dr & Monroe St and Michigan Ave & Oak St.



Slide 8: Strategic Marketing Recommendations (Act)

01

INTRODUCE A WEEKEND-ONLY PASS

Casual riders peak heavily on Saturday and Sunday. Designing a specific weekend-only annual membership will capture these recreational riders who do not need a weekday commute option.

02

PROMOTE FINANCIAL SAVINGS

Casual riders average 22.5 minutes per trip (double that of members). Run campaigns proving that purchasing an annual membership saves money compared to buying multiple single-ride passes.

03

STATION-SPECIFIC MARKETING

Focus conversion ads, QR code discounts, and physical marketing stands directly around the top 10 casual rider starting stations (e.g., Navy Pier, DuSable Lake Shore Dr, Millennium Park).

04

SUMMER CAMPAIGN WINDOW

Casual rides skyrocket in June, July, and August, dropping to near-zero in winter. Focus the marketing budget heavily during these summer months when engagement and interest are at their peak.

Thank You!

Questions & Answers | Google Data Analytics Capstone Case Study

Author: Sergey Kasatov

Google Data Analytics Professional Certificate Capstone

Date: July 2026
